0.1 The complete quaternion correction fibre in the $\mathrm{SL}_2(\mathbf{F}_3)$ Ferrari target

Statement E.59 (Proved). Every retained CM matrix pair survives the complete $\mathrm{SL}_2(\mathbf{F}_3)$ Ferrari test. *Retain the actual groups B, F, N_B, G_6 and the 24 indexed, multiplier-one G_6 points of E.17. There is a specified finite Ferrari quotient*

$$\pi_{24} : B_4 \twoheadrightarrow T_{24}, \qquad |T_{24}| = 2^{29}3^4 = 43\,486\,543\,872, \tag{E.59.1}$$

obtained from every epimorphism kernel of the free Ferrari group $\langle \xi_1, \xi_2 \rangle$ onto $\mathrm{SL}_2(\mathbf{F}_3)$. Let $\rho_{24} : B \twoheadrightarrow G_{24}$ be the actual induced source formed from all six E.17 evaluations and all six B/F cosets. Then

$$\begin{gathered} |G_{24}| = 2^{26}3^6 = 48\,922\,361\,856, \qquad P = \rho_{24}(N_B), \quad |P| = 2^{20}3^4, \\ C = \rho_{24}(N'_B) = \rho_{24}(N'_B N_B^6) = \rho_{24}(F' \cap N'_B N_B^6) \simeq Q_8^3. \end{gathered} \tag{E.59.2}$$

The isomorphism in the last line is the product of three actual evaluation-coordinate homomorphisms described below.

Writing $K_6 = N'_B N_B^6$, the common source is exactly

$$D_{24} = B/(K_6 \cap \ker \rho_{24}) \simeq G_6 \times_{G_{24}/C} G_{24}, \qquad \ker(D_{24} \to G_6) = C. \tag{E.59.3}$$

For each one of the 24 retained normalized words $f_i \in F'$, the whole finite-value fibre of actual F' words over its G_6 value is $f_i C$, with 512 elements. In every one of these fibres, the exact counts at multiplier one are

literal finite conditions	*number of values*
$(sf)^2 = (f^{-1}r)^3 = 1$ *in* D_{24}	4
the preceding conditions and the full target pentagon	2
also all three four-strand braid relations	2.

(E.59.4)

All four first-row points generate the actual three-strand source D_{24}. Both last-row points generate the whole target T_{24} through the specified four-strand braid map and fix the literal three- and four-strand full twists. Every value has a retained literal commutator-word preimage in F'.

The 24 base points have the 24 distinct admissible CM matrix pairs proved in E.1 and retained in E.17, including all 12 extra pairs. Thus each of those matrix fibres has two values passing this complete finite test. The counts in (E.59.4) *are counts in the entire 512-element fibre over each specified G_6 point. They do not enumerate other affine G_6 points or other multiplier residues in the same CM matrix fibre, and do not assert a compatible full profinite pentagon lift.*

Proof. We construct the target and its faithful source, determine the whole correction image, and then check the exact finite equations and generation.

The actual universal target. Put $S = \mathrm{SL}_2(\mathbf{F}_3)$. Enumerate its 24 determinant-one matrices and the 384 ordered generating pairs. Conjugation by $\mathrm{GL}_2(\mathbf{F}_3)$ supplies 24 different automorphisms. To verify completeness without assuming an abstract automorphism-group identification, choose one generating pair of S, retain a Cayley spanning tree, and test the homomorphism given by every proposed generating-pair image on every Cayley edge. Exactly the same 24 maps pass. Thus these are all of $\mathrm{Aut}(S)$.

Two epimorphisms from $L = \langle \xi_1, \xi_2 \rangle$ onto S have the same kernel exactly when an automorphism of S relates them: the reverse map is the actual isomorphism induced from

their common quotient $L/\ker\varphi$. There are therefore 16 epimorphism kernels. Use the lexicographic generating-pair representative (g_j, h_j) of each orbit and the actual evaluation map

$$L \longrightarrow S^{16}, \qquad \xi_1 \longmapsto (g_j)_j, \quad \xi_2 \longmapsto (h_j)_j. \tag{E.59.5}$$

Its image K is retained as that subgroup; no independence of the 16 factors is presumed.

The exact Ferrari Nielsen substitutions are

$$a : (g,h) \longmapsto (g, hg), \qquad b : (g,h) \longmapsto (gh^{-1}, h). \tag{E.59.6}$$

For each representative, the resulting pair determines its new kernel and the unique automorphism of S relating it to the chosen new representative. This gives a permutation p and automorphisms α_j in the retained order. A general element of the containing holomorph is written (p, α, k), with $k_j \in S$. Its product is

$$(p,\alpha,k)(q,\beta,l) = (q \circ p, (\alpha_j \beta_{p(j)})_j, (k_j \alpha_j(l_{p(j)}))_j). \tag{E.59.7}$$

It has the faithful right permutation action on the 384 points (j,t), $t \in S$, given by

$$(j,t) \longmapsto (p(j), \alpha_j^{-1}(k_j^{-1} t)). \tag{E.59.8}$$

Faithfulness follows first from the factor label. The image of 1 then gives the constant $\alpha_j^{-1}(k_j^{-1})$; multiplying it off the remaining values recovers α_j^{-1}, and hence k_j. Substitution in (E.59.7) verifies composition in this right-action convention.

Take the three braid generators to be $a, b, c = \xi_1 a$ in this model. All braid relations, the two literal Ferrari words, and all four conjugation formulas are checked. In particular

$$\xi_1 = ca^{-1}, \qquad \xi_2 = abac^{-1}b^{-1}a^{-1}, \tag{E.59.9}$$

with conjugation by a fixing ξ_1 and sending ξ_2 to $\xi_2\xi_1$, and conjugation by b sending ξ_1 to $\xi_1\xi_2^{-1}$ and fixing ξ_2. These checks prove that the construction is a quotient of the actual Ferrari braid group. Exact faithful permutation calculations give

$$|K| = 2^{14}3^2, \qquad |\langle a,b\rangle| = 2^{15}3^2, \qquad |\langle a,b,c\rangle| = 2^{29}3^4. \tag{E.59.10}$$

The first factor is normal under the displayed Nielsen action, and the coordinates of $\langle a,b\rangle$ have $k = 1$. Thus their intersection is trivial, and the product of the first two orders is the last order. The program independently computes all three.

The complete six-evaluation source. Use the positive pure braid words A_{ij} of E.17 and $\omega = (ab)^3$. The six pairs are, literally,

$$\begin{gathered}(A_{12}, A_{23}A_{24}), \quad (A_{13}A_{23}, A_{34}), \quad (A_{23}, A_{34}),\\ (A_{12}A_{13}, A_{24}A_{34}), \quad (A_{12}, A_{23}), \quad (\omega, A_{34}).\end{gathered} \tag{E.59.11}$$

Let $\theta_i : F \to T_{24}$ denote the corresponding actual evaluation. For the saved section $q \mapsto b_q$ of $B/F = S_3$, define the component of $\rho_{24}(w)$ at (q,i) to be

$$\theta_i(b_q w b_{qw}^{-1}), \tag{E.59.12}$$

together with the coset permutation $q \mapsto qw$. The cocycle identity for these based words proves that this is a homomorphism. At the identity coset its restriction to F is exactly θ_i.

The resulting faithful permutation action has degree $6 \cdot 6 \cdot 384 = 13\,824$. It compresses to degree 1956 by discarding only duplicate transitive B-sets. Each discarded orbit has

an explicit bijection to its retained copy intertwining both s and r, so the compressed and original maps have the identical kernel. The retained orbit sizes are

$$48, 24, 432, 384, 96, 432, 96, 12, 432. \tag{E.59.13}$$

The actual original generator multiplication and both torsion relations are checked before compression. Complete Schreier chains then give the first two orders in (E.59.2), using the literal 97 Bass–Serre generators of N_B for the subgroup image P. Their index is $576 = [B : N_B]$. Consequently $\ker \rho_{24} \subset N_B$: indeed $[B : N_B \ker \rho_{24}] = 576$ already equals $[B : N_B]$. The specified matrix quotient $q : B \to Q$ therefore factors through G_{24} with its original generator values.

The whole correction image and its actual quaternion maps. Evaluate all 4656 pair commutators of the 97 retained basis elements. There are 158 distinct nonidentity values. Their generated subgroup has order 512. In the zero-based basis numbering, six generating commutators are

$$[n_0, n_1],\ [n_0, n_4],\ [n_0, n_5],\ [n_0, n_6],\ [n_8, n_{11}],\ [n_8, n_{13}]. \tag{E.59.14}$$

Every original pair commutator is in this subgroup, and conjugating these six generators by each original n_i, and separately by s, r, stays in the same subgroup. This proves that it is exactly $C = \rho_{24}(N'_B)$, with the asserted normality in G_{24}. An independent complete derived-subgroup Schreier calculation also gives order 512.

For each of the 36 evaluation coordinates of (E.59.12), restriction to C is a homomorphism, because $C \subset \rho_{24}(F)$. The three retained coordinate projections with zero-based indices $(q, i) = (0, 0), (0, 1), (0, 3)$ have image of order eight, each with exactly one involution and with explicit generators i, j satisfying

$$i^2 = j^2 = -1, \qquad ij = (-1)ji. \tag{E.59.15}$$

The full multiplication table and the selected coordinate indices are saved. Each image is therefore the explicitly presented quaternion group. The combined map separates all 512 elements of C, and its target has order $8^3 = 512$. It is consequently an isomorphism $C \to Q_8^3$ through those actual homomorphisms. In particular its centre has order eight and its exponent is four.

Every one of the 97 values $\rho_{24}(n_i^6)$ lies in C. Since P/C is abelian and generated by these basis images, its exponent divides six. Hence every sixth power in P lies in C, proving $\rho_{24}(K_6) = C$. On the other hand the six generators (E.59.14) have literal preimages in $N'_B \subset K_6 \cap F'$. This gives both inclusions in $\rho_{24}(K_6 \cap F') = C$. A breadth-first enumeration of the 512 elements retains a parent and a generator label for every element, hence an actual N'_B word preimage of each one.

The map $G_6 \to G_{24}/C$ is induced from these exact source maps. For any proposed pair in the right side of (E.59.3), lift its G_6 coordinate to B. The discrepancy of its second coordinate lies in $C = \rho_{24}(K_6)$ and is corrected by an actual K_6 element. This proves the fibre-product isomorphism and its kernel. The same argument with the saved N'_B words proves that every point of $f_i C$ has an actual F' lift retaining the specified G_6 value. Conversely any such lift differs by $K_6 \cap F'$, whose entire image was just computed. This proves completeness of the finite correction fibre.

The exact finite equations. For every one of these 512 corrections, evaluate the normalized three-strand images

$$S_f = sf, \qquad R_f = f^{-1}r. \tag{E.59.16}$$

Their product is the original $\tau = sr$. The original G_6 values already satisfy $S_f^2 = R_f^3 = 1$. Testing these identities in all 36 coordinates of (E.59.12) therefore tests them in the faithful common source D_{24}. Exactly four corrections in every fibre pass.

Write $v_j = \theta_j(f)$ for the six target values in (E.59.11). The full group pentagon, with the retained conventions, is the literal equality

$$v_1 v_2 = v_3 v_4 v_5. \tag{E.59.17}$$

The associated four-strand generator images are

$$a, \qquad v_5^{-1} b v_5, \qquad v_6^{-1} c v_6. \tag{E.59.18}$$

For all four gentle points, the program tests (E.59.17) and each of the three Artin braid relations for (E.59.18). Exactly two pass all of them, in every one of the 24 fibres. These are exhaustive group multiplications in (E.59.7), including the nonabelian quaternion corrections. No linearization, selected correction, or omitted central term is used. The saved output gives every passing index and both complete literal commutator-word lifts in each fibre.

Source and target generation. For each of the four gentle points in each fibre, the complete permutation calculation on the 1956 retained source points gives

$$|\langle S_f, R_f \rangle| = |G_{24}|. \tag{E.59.19}$$

Thus the corrected map is onto G_{24} and is already onto G_6 through its retained old shadow. These two surjections give a surjection onto the actual common source, as follows. Their common composite to Q is $\mathrm{Inn}(g_i) \circ q$, where $g_i = q(x)^{h_i}$ is the actual integral cusp gauge retained for the i-th old point. This equality is checked on the literal s, r images. In particular its kernel is N_B. The image of N_B is therefore a subdirect product of the two actual N_B images. The G_6 image is abelian. By Goursat, their common quotient is abelian, and so kills the commutator $P' = C$ on the G_{24} side. The corrected image therefore contains the entire kernel C of (E.59.3) and maps onto G_6. It is all of D_{24}. This proves the complete three-strand generation assertion through the specified maps.

For each of the two pentagon points, a separate exact Schreier calculation on the 384 target points gives the order $2^{29}3^4$ for the three images in (E.59.18). They belong to T_{24}, so they generate precisely T_{24}. At all 48 pentagon points, literal multiplication also verifies that $(ab)^3$ and $(abc)^4$ retain their original target values. Every chain retains the original two or three generators and literal-word directed acyclic graph. All final Schreier relations are sifted, and the independent Python array reader re-evaluates every saved strong-generator word. Thus the order equalities are complete finite generation certificates.

Finally, retain the actual ungauged matrix frames A_i, β_i of each E.17 point. Its gauge g_i changes the three frames to $(g_i A_i, g_i \beta_i, g_i)$. Consequently the CM coordinates are exactly

$$(g_i A_i)^{-1} g_i = A_i^{-1}, \qquad (g_i \beta_i)^{-1} g_i = \beta_i^{-1}. \tag{E.59.20}$$

The separate literal matrix check retains the nontrivial gauges, the unique E.1 representatives with mod-four value in V_4, and the positive rational three-torsion line. It obtains 24 distinct pairs and exactly 12 unequal sign pairs. E.1's finite-centralizer calculation shows that reduction to the same G_6 point preserves those matrix coordinates under the current corrections. This proves the stated CM coverage through the actual frames, including the original $12 + 12$ division.

The target and source programs in the code directory are `a_cm_sl23_model.py` and `a_cm_sl23_induced.py`. The complete fibres are computed in `a_cm_sl23_corrections.py`;

generation is checked in `a_cm_sl23_generation.py`. The full twists are checked in `a_cm_sl23_periodic.py`. The actual CM frames are checked in `a_cm_sl23_cm_labels.py`. All outputs and exact certificate dependencies are retained. The initial word-only probe is retained separately with its explicitly narrower scope: its 23 failing representative words do not exclude their finite fibres, as the exhaustive correction calculation proves. □